

LogiCraft Operations Portal

Complete Master Presentation & Technical Reference Guide

Stack: Python 3.13 • Django 5.x • Leaflet GIS • Bootstrap 5

Environment: Localhost (127.0.0.1:8000)

Date: September 2026

Project Mission Statement

LogiCraft is an integrated fleet and logistics management system that combines **real-time GIS telematics tracking**, **freight consignment lifecycle management**, **regional warehouse capacity planning**, and **multi-modal passenger transit ticketing** into a unified, enterprise-ready platform.

1. Staff Roles & Login Credentials

The system enforces Role-Based Access Control (RBAC) via a custom user model. All roles can be tested via the login portal at <http://127.0.0.1:8000/users/login/>.

Role	Username	Password	Role Badge	Permissions & Scope
System Administrator	admin	admin123	ADMIN	Full system governance, user permissions, master fleet & warehouse registries, Django Admin.
Logistics Manager	manager1	manager123	MANAGER	Freight dispatch, vehicle fleet allocation, shipment waybill generation, inventory auditing.
Fleet Driver	driver_rajesh	driver123	DRIVER	Dedicated telemetry console, assigned trip checkpoints, turn-by-turn route overview.
Public Customer	(No Login Required)	(None)	PUBLIC	Public consignment tracking portal via tracking number (e.g. LC-2026-DELMUM).

2. Complete System URLs Directory

Feature / Page	Direct URL	Description & Key Highlights
Portal Login	/users/login/	Role-based authentication gateway with secure session tokens and redirection.
Executive Dashboard	/	Live operations cockpit: 4 KPI metric cards, fleet health stats, active transit routes.
Live GIS Telematics Map	/fleet/map/	Flagship Feature: Interactive Leaflet map, Esri World Streets / OSM / Satellite layer switcher, fly-to sidebar.
Fleet Ledger	/fleet/	Complete vehicle registry with status tags (Available, On Trip, Maintenance) and odometers.
Vehicle Telemetry View	/fleet/vehicle/1/	Tata Signa hauler telemetry: GPS coordinates, payload capacity, maintenance service logs.

Feature / Page	Direct URL	Description & Key Highlights
Register Vehicle	/fleet/vehicle/add/	Form to onboard heavy trucks, delivery vans, and transit coaches with payload configs.
Export Fleet CSV	/fleet/export-csv/	Instant streaming CSV download of entire vehicle fleet for Excel/PowerBI reporting.
Freight Shipments	/shipments/	Consignment ledger tracking weights, freight costs, origin/destinations, and status pills.
Create Consignment	/shipments/new/	Auto-generates unique AWB tracking codes, assigns carriers, drivers, and cargo specs.
Shipment Detail & Map	/shipments/1/	Progress timeline milestones (Booked → Picked Up → In Transit) and mini GIS route line.
Printable Waybill	/shipments/1/waybill/	Official consignment waybill formatted for thermal printing and dispatch sign-off.
Public Parcel Tracker	/shipments/track/	Customer tracking page: live truck GPS pin, route polyline, and checkpoint history.
Regional Warehouses	/warehouse/	Storage hubs (Delhi, Navi Mumbai, Bengaluru) with square-footage utilization metrics.
Warehouse Inventory Detail	/warehouse/1/	Delhi Central Hub inventory items (Smart TVs, Apparel), unit weights, and hub manager contact.
Transit Bus Routes	/transport/	Intercity schedules (Delhi → Agra, Mumbai → Pune) with departure timings, fares, and coaches.
Passenger Bookings	/bookings/	Passenger manifests, assigned seat numbers (e.g. 12A), travel dates, and fares collected.
Book Ticket Form	/bookings/book/1/	Real-time seat reservation form decrementing available bus seats automatically.
Django Administration	/admin/	Direct ORM database administration console for system maintainers and DBAs.

3. Technical Architecture & Design Decisions

Backend & ORM Design Built with Python 3.13 & Django 5.x using clean Model-View-Template separation. Data integrity is enforced using foreign key cascades, unique tracking number generators, and decimal-precision financial calculations.	GIS & Mapping Strategy Integrated Leaflet.js with Esri World Street Map , OpenStreetMap , and Esri Satellite Imagery . Completely eliminates third-party API keys, billing accounts, and avoids OSM volunteer server 403 blocks.
Role-Based Access (RBAC) Extended <code>AbstractUser</code> with a custom <code>Role</code> enumeration (<code>ADMIN</code> , <code>LOGISTICS_MANAGER</code> , <code>DRIVER</code>). Decorators restrict critical endpoints like vehicle onboarding and CSV exports.	Reporting & Data Export Native HTTP streaming using Python's <code>csv</code> writer generates RFC-4180 compliant CSV files with live metadata headers, allowing instant data extraction into BI tooling.

4. Step-by-Step Presentation Script (How to Demo)

Follow this curated 6-minute flow to deliver a confident and structured presentation.

Step 1: Introduction & Role-Based Access Control Action: Open <code>http://127.0.0.1:8000/users/login/</code> and log in as <code>admin / admin123</code> . <i>"LogiCraft is an enterprise logistics portal built in Django 5. It enforces strict role-based access control where system admins manage infrastructure, logistics managers coordinate dispatches, and fleet drivers have dedicated mobile-friendly views."</i>	Duration: 1.0 Min
Step 2: Executive Dashboard KPIs Action: Navigate to the home dashboard (<code>http://127.0.0.1:8000/</code>). <i>"The dashboard gives operations directors an immediate bird's-eye view: active vehicles on the road, high-priority cargo consignments, storage footprint across hubs, and active passenger transit routes across the country."</i>	Duration: 1.0 Min
Step 3: Live GIS Telematics Map (The Showstopper) Action: Navigate to <code>http://127.0.0.1:8000/fleet/map/</code> . Demonstrate: 1) Click the top-right layer switch to toggle between <i>World Street Map</i> and <i>Satellite Imagery</i> . 2) Click any asset in the right sidebar — the map smoothly flies to and centers on the marker. 3) Point out the blue dashed Delhi-to-Mumbai freight corridor. <i>"This is our GIS telematics engine. Rather than relying on costly third-party map APIs with rate limits, we integrated high-performance Esri and OpenStreetMap layers. We plot real-time vehicle GPS coordinates, interstate freight corridors, and warehouse hubs with smooth fly-to animations."</i>	Duration: 2.0 Min

Step 4: Freight Lifecycle & Public Tracking

Duration: 1.5 Min

Action: Open shipment LC-2026-DELMUM at /shipments/1/. Show the milestone timeline and click **Print Waybill**. Then open an Incognito window to /shipments/track/?tracking_number=LC-2026-DELMUM.

"Every shipment maintains a verifiable audit trail with milestone checkpoints. Consignors can instantly generate standard freight waybills, while end-customers can track their parcels in real time without needing an account."

Step 5: Multi-Modal Transit & Bus Ticketing

Duration: 0.5 Min

Action: Open /transport/ and /bookings/. Show the Volvo coach intercity routes and passenger seat allocations.

"LogiCraft also supports passenger operations. Intercity coaches have scheduled routes, live seat availability tracking, and automated passenger ticket issuance."

5. Evaluator Q&A Cheat Sheet (Common Viva Questions)

Q: How does the map render live locations without paid Google Maps APIs?

Answer: We use Leaflet.js with **Esri World Street Map** and **OpenStreetMap** raster tile servers. Leaflet downloads raster tiles asynchronously over HTTPS with zero billing dependencies, while Django ORM serializes model coordinates to GeoJSON-compatible objects on the client.

Q: How are coordinates and vehicle locations updated?

Answer: The `Vehicle` model stores float fields for `latitude` and `longitude` alongside timestamps and current location descriptions. In production, IoT telematics devices (OBD-II / GPS transponders) send telemetry updates to an internal REST webhook endpoint which updates these coordinates.

Q: How is data protected between different user roles?

Answer: Through Django's built-in session framework coupled with custom view decorators (e.g. `@login_required` and role verification checks). Views evaluate `request.user.role` before granting access to sensitive views like fleet modifications or financial waybills.

Q: How does the system scale to handle large datasets?

Answer: We use Django database query optimizations like `select_related` and `prefetch_related` to eliminate N+1 query overhead across relationships (e.g. Driver-to-Vehicle, Shipment-to-Warehouse). CSV exports utilize Python generators to stream rows without consuming server RAM.

Golden Presentation Strategy

Before stepping into the presentation: Open the 4 primary tabs in your browser in advance:

1) **Dashboard** (/), 2) **GIS Map** (/fleet/map/), 3) **Shipment Waybill** (/shipments/1/), and 4) **Public Tracking** (/shipments/track/).

Seamlessly flipping between these tabs demonstrates high product Polish and zero lag!